A Appendix

A.1 Estimating the Mutual Information with InfoNCE

By optimizing InfoNCE, the CPC loss we defined in Equation 4, we are maximizing the mutual information between c_t and z_{t+k} (which is bounded by the MI between c_t and x_{t+k}). This can be shown as follows.

As already shown in Section 2.3, the optimal value for $f(x_{t+k}, c_t)$ is given by $\frac{p(x_{t+k}|c_t)}{p(x_{t+k})}$. Inserting this back in to Equation 4 and splitting X into the positive example and the negative examples X_{neg} results in:

$$\mathcal{L}_{\text{N}}^{\text{opt}} = -\mathop{\mathbb{E}}_{X} \log \left[\frac{\frac{p(x_{t+k}|c_t)}{p(x_{t+k})}}{\frac{p(x_{t+k}|c_t)}{p(x_{t+k})} + \sum_{x_j \in X_{\text{neg}}} \frac{p(x_j|c_t)}{p(x_j)}} \right] \tag{6}$$

$$= \mathop{\mathbb{E}}_{X} \log \left[1 + \frac{p(x_{t+k})}{p(x_{t+k}|c_t)} \sum_{x_j \in X_{\text{neg}}} \frac{p(x_j|c_t)}{p(x_j)} \right] \tag{7}$$

$$\approx \mathop{\mathbb{E}}_{X} \log \left[1 + \frac{p(x_{t+k})}{p(x_{t+k}|c_t)} (N-1) \mathop{\mathbb{E}}_{x_j} \frac{p(x_j|c_t)}{p(x_j)} \right] \tag{8}$$

$$= \mathop{\mathbb{E}}_{X} \log \left[1 + \frac{p(x_{t+k})}{p(x_{t+k}|c_t)} (N-1) \right] \tag{9}$$

$$\geq \mathop{\mathbb{E}}_{X} \log \left[\frac{p(x_{t+k})}{p(x_{t+k}|c_t)} N \right] \tag{10}$$

$$= -I(x_{t+k}, c_t) + \log(N), \tag{11}$$

Therefore, $I(x_{t+k}, c_t) \geq \log(N) - \mathcal{L}_{\text{N}}^{\text{opt}}$. This trivially also holds for other f that obtain a worse (higher) $\mathcal{L}_{\text{N}}$. Equation 8 quickly becomes more accurate as N increases. At the same time $\log(N) - \mathcal{L}_{\text{N}}$ also increases, so it's useful to use large values of N.

InfoNCE is also related to MINE [54]. Without loss of generality let's write $f(x, c) = e^{F(x,c)}$, then

$$\mathop{\mathbb{E}}_{X} \left[\log \frac{f(x,c)}{\sum_{x_j \in X} f(x_j, c)} \right] = \mathop{\mathbb{E}}_{(x,c)} \Big[F(x,c) \Big] - \mathop{\mathbb{E}}_{(x,c)} \Big[\log \sum_{x_j \in X} e^{F(x_j,c)} \Big] \tag{12}$$

$$= \mathop{\mathbb{E}}_{(x,c)} \Big[F(x,c) \Big] - \mathop{\mathbb{E}}_{(x,c)} \Big[\log \Big(e^{F(x,c)} + \sum_{x_j \in X_{\text{neg}}} e^{F(x_j,c)} \Big) \Big] \tag{13}$$

$$\leq \mathop{\mathbb{E}}_{(x,c)} \Big[F(x,c) \Big] - \mathop{\mathbb{E}}_{c} \Big[\log \sum_{x_j \in X_{\text{neg}}} e^{F(x_j,c)} \Big] \tag{14}$$

$$= \mathop{\mathbb{E}}_{(x,c)} \Big[F(x,c) \Big] - \mathop{\mathbb{E}}_{c} \Big[\log \frac{1}{N-1} \sum_{x_j \in X_{\text{neg}}} e^{F(x_j,c)} + \log(N-1) \Big] \tag{15}$$

is equivalent to the MINE estimator (up to a constant). So we maximize a lower bound on this estimator. We found that using MINE directly gave identical performance when the task was non-trivial, but became very unstable if the target was easy to predict from the context (e.g., when predicting a single step in the future and the target overlaps with the context).